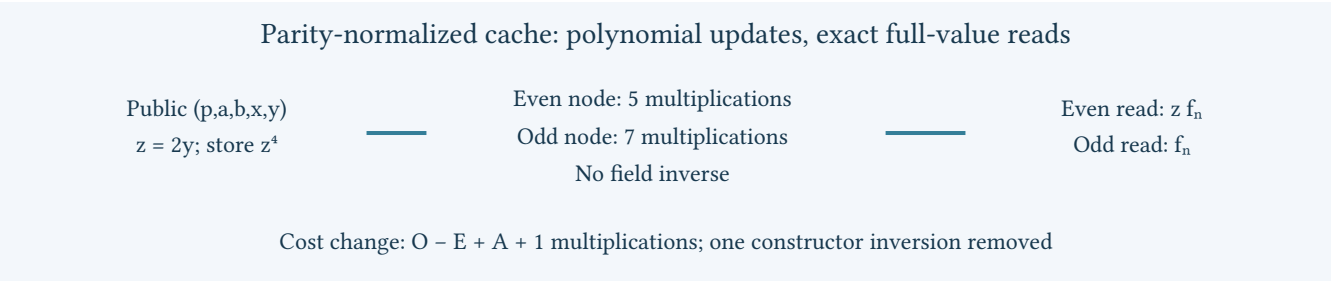
Division values without a field inverse

T-1825 · 2026-11-07 · parity-normalized native cache

The value-only division cache now has an inversion-free route. It stores f(n)=W(n)/(2y) for even n and W(n) for odd n, with polynomial-only initialization. The public API restores exact W(n) values. The general dyadic quotient source integrates this cache with unchanged seed, resultant, window and sign logic.



Derived recurrence. Set z=2y. Even: $f(2m)=f(m)[f(m+2)f(m-1)^2-f(m-2)f(m+1)^2]$. For odd indices, let $L=f(m+2)f(m)^3$ and $R=f(m-1)f(m+1)^3$. Use z^4L-R when m is even, $L-z^4R$ when m is odd. These follow by substituting parity factors into the standard division-polynomial identities. No field inverse or exponentiation is called in the new cache.

Exact check	Count	Coverage
Full division-value equalities	144,040	1,108 points × 130 indices
Antipodal identities	144,040	Same points and indices
Repeated reads / cache sizes	144,040	No retained-index growth
Integrated source points	1,108	Reference, half-products, signs
Identity-crossing windows	24	Prepared public windows
Additional recovery targets	617	5,929 calls; zero unresolved
Inherited public comparisons	63	511 source calls

Boundary controls. Eight prime fields: 5,7,17,41,97,193,257,509. All nonzero-y curve points are checked at indices0–127 and the two largest u64 indices. Domain, noncanonical-input, singular-curve, off-curve and zero-y rejections pass. Maximum indices test recursion bounds, not cryptographic-size fields. Normal metadata build has five dependencies, excluding the old value cache and derivative source.

Exact cost tradeoff. Let E and O count new nonbase even and odd recurrence nodes, A all even public reads. The old recurrence uses 6E+6O multiplications; this one uses 5E+7O+A. Constructor changes add one multiplication and remove one inverse. Thus delta is O−E+A+1 multiplications and −1 inversion. These are expression counts, not timings. Under an I/M cost model, this route is preferred only when the saved inversion exceeds the extra multiplication cost, with other costs assessed separately.

Scope and evidence. Prime/nonzero-y and word-size conditions remain. Source coprimality, r≠p and lane-cap conditions remain. Geometry and polynomial operations retain their own inversions. Original4,563-case cohort unchanged; no cloud or timing run. All focused native builds/assertions pass; full repository Cargo suite not rerun. PCC-20261107-5b8ec1 is unsigned producer evidence. See DERIVATION.md and receipts for replay/hashes.

Sources and next step. Mathlib EDS documentation gives standard even/odd identities; Lauter–Stange2008 provides inherited normalization context. KB unavailable. No novelty or Lean-verified claim. Next compute the public request-index DAG cost and combine it with remaining source arithmetic before backend selection.